มหาวิทยาลัยมหิดล
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ITDS261 Introduction to Software Engineering

Lecture 2 – Software Process Models

Photo: https://www.flickr.com/photos/ninara/38707088530

Topics

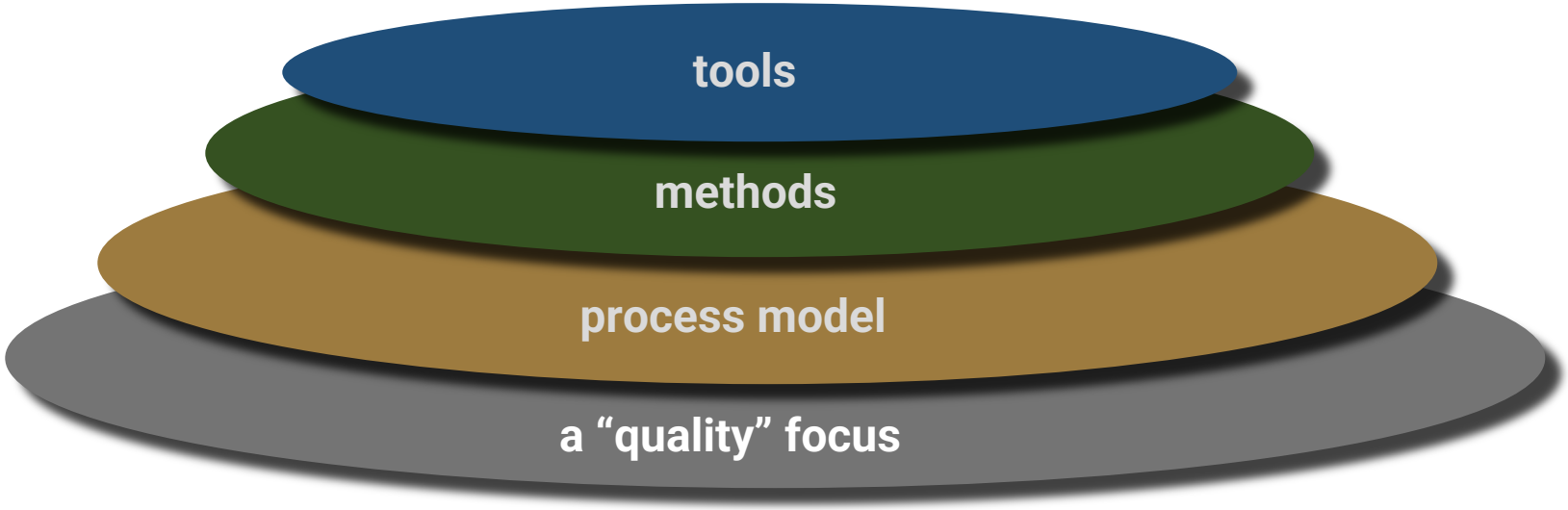
- Software process model definition
- Well-known software process models: Conventional vs. Modern
- Software process improvement

Case Study: CrowdStrike Outage

A bug in the Content Validator, part of CrowdStrike's deployment process, allowed an error through, the security company revealed on July 24.

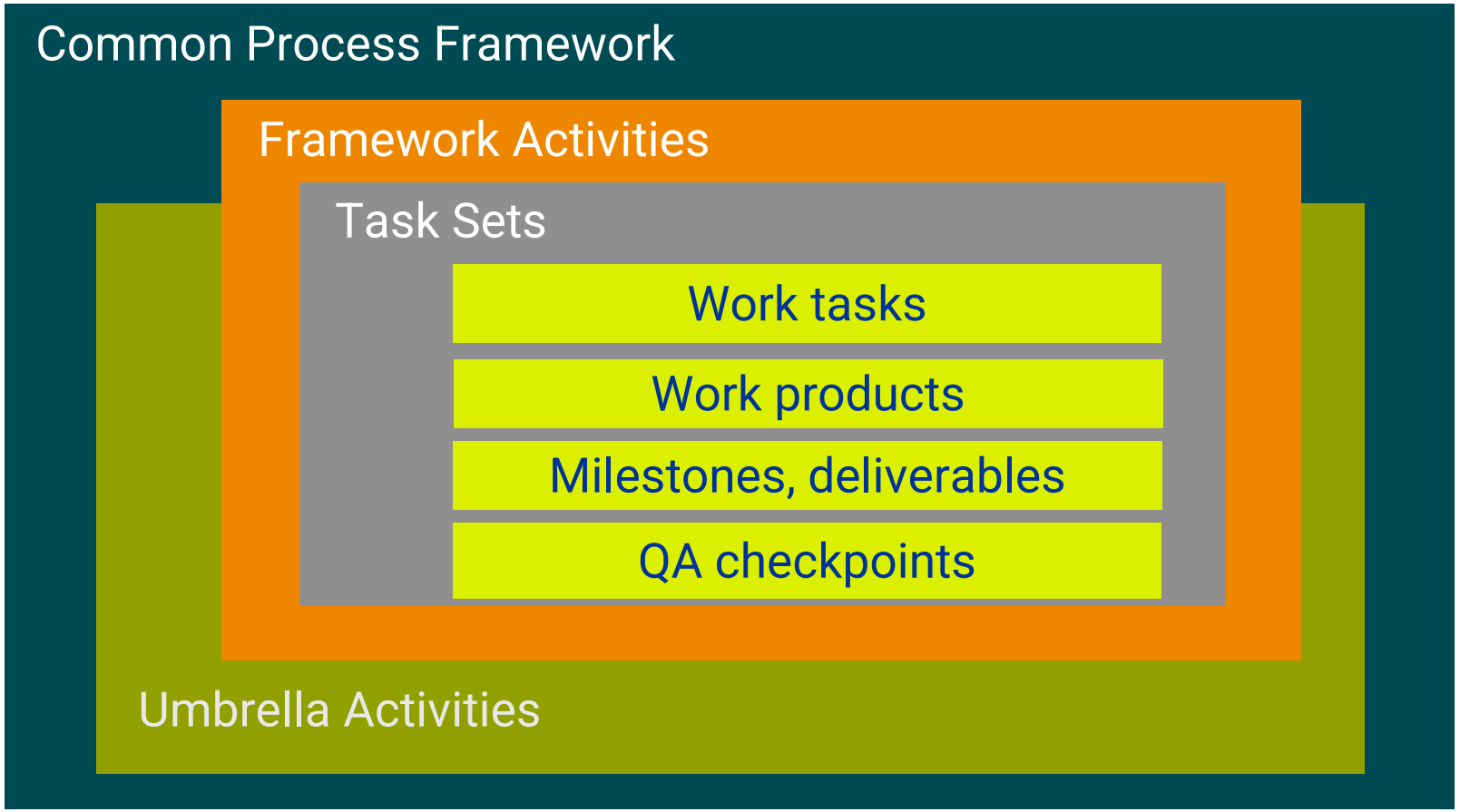
References: https://www.youtube.com/watch?v=okwL2NQi_G4,
Techrepublic (2024) CrowdStrike: Buggy Validator Started Massive Outage

A Layered Technology

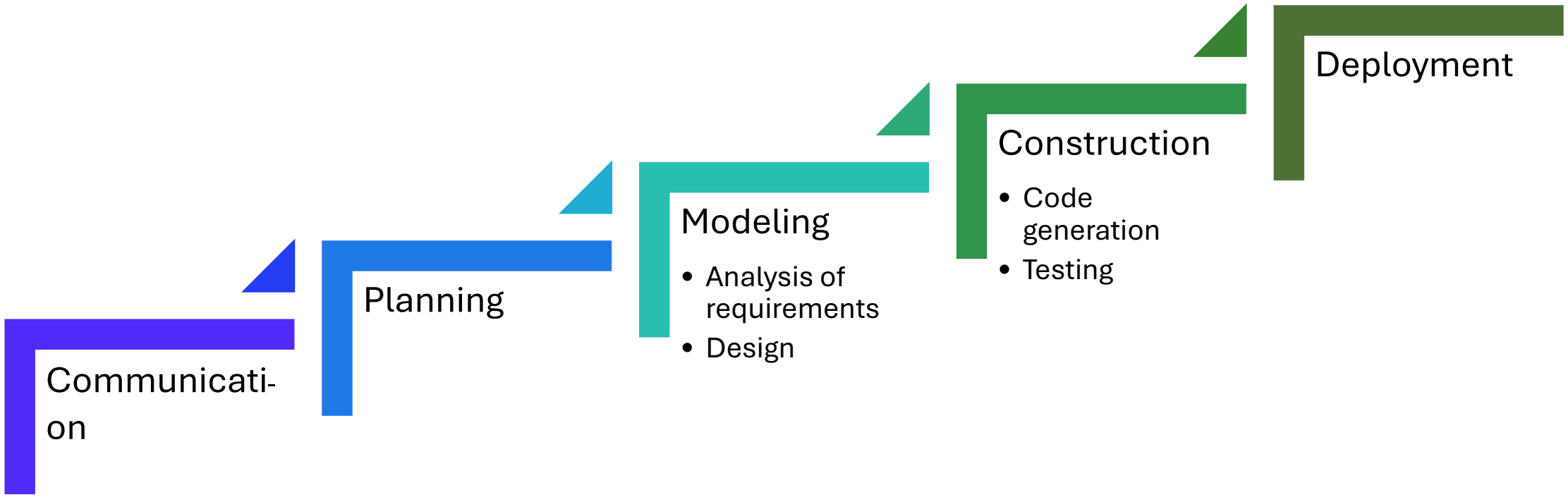


Software Engineering

A Process Framework



Framework Activities



Umbrella Activities

- Software project management
- Formal technical reviews
- Software quality assurance
- Software configuration management
- Work product preparation and production
- Reusability management
- Measurement
- Risk management

Process Models

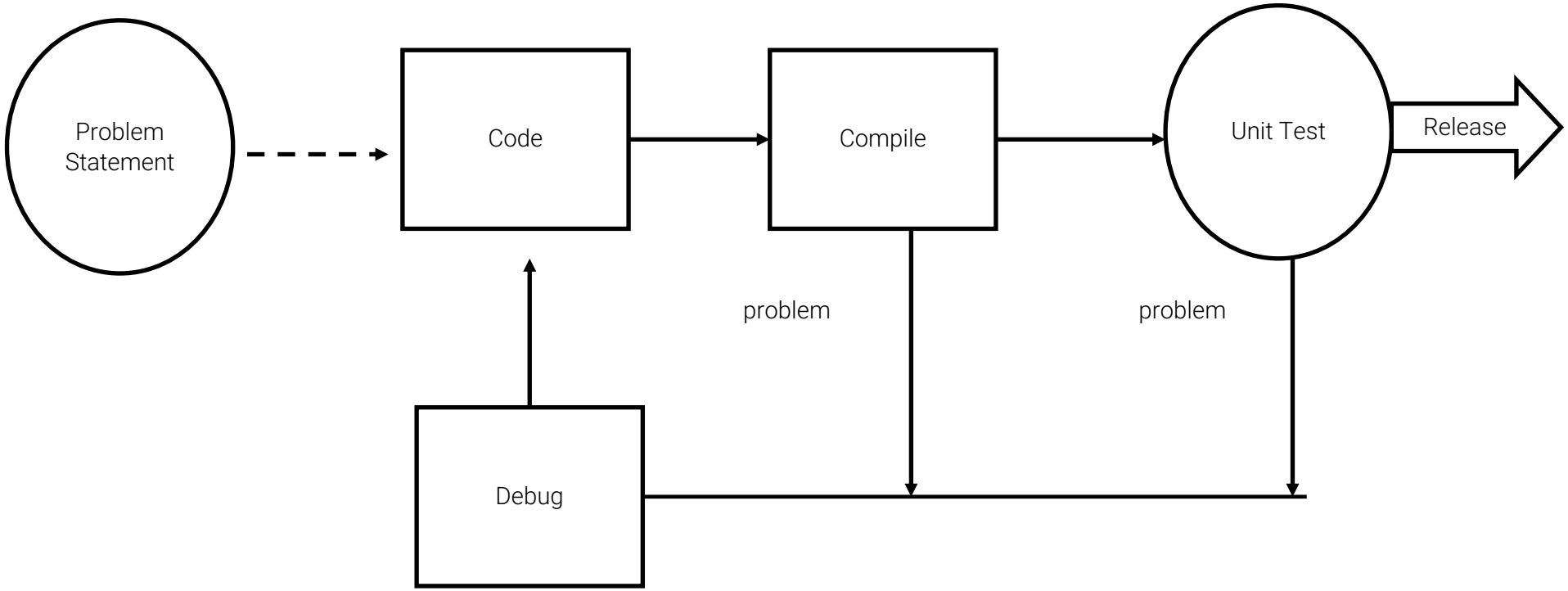
- Process models are standardised frameworks
- Provide guidance for a systematic coordination and controlling of the tasks and of the personnel who performs the tasks
- Describe the sequence of activities, roles, artifacts, and workflows involved in the development and maintenance of a software system
  - To guide software teams in planning, executing, and managing projects
  - To ensure consistency and quality of work product
  - To facilitate communication and coordination among stakeholders

*** coordination, tasks, people ***

Process Models (Cont.)

- What to define:
  - Set of tasks that need to be performed
  - Input and output of those tasks
  - Pre-conditions and post-conditions of those tasks
  - Sequence of flow of the tasks
  - Personnel who perform the tasks
- Examples:
  - Code and Fix
  - Waterfall
  - Incremental
  - Spiral
  - Agile

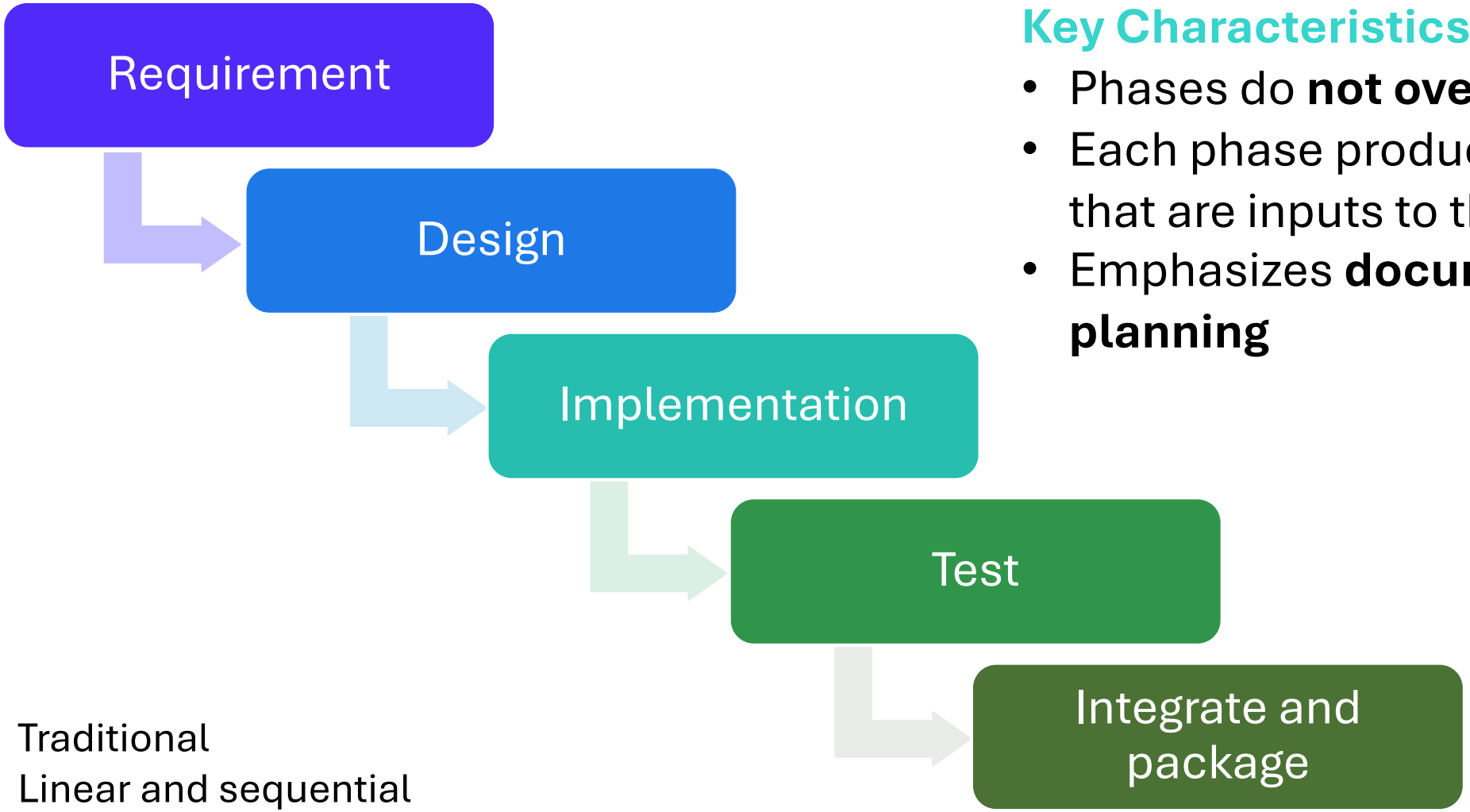
Code and Fix Model



1. Most people performs and follow this process, but unfortunately some *skips* unit testing or debugging
2. Also, some proceeds *without* clearly understanding the "problem statement" ---- which is requirement

Waterfall Model

Software Development Life Cycle (SDLC)



Key Characteristics:

- Phases do **not overlap**.
- Each phase produces deliverables that are inputs to the next.
- Emphasizes **documentation** and **planning**

Traditional
Linear and sequential

Pros and Cons: Waterfall Model

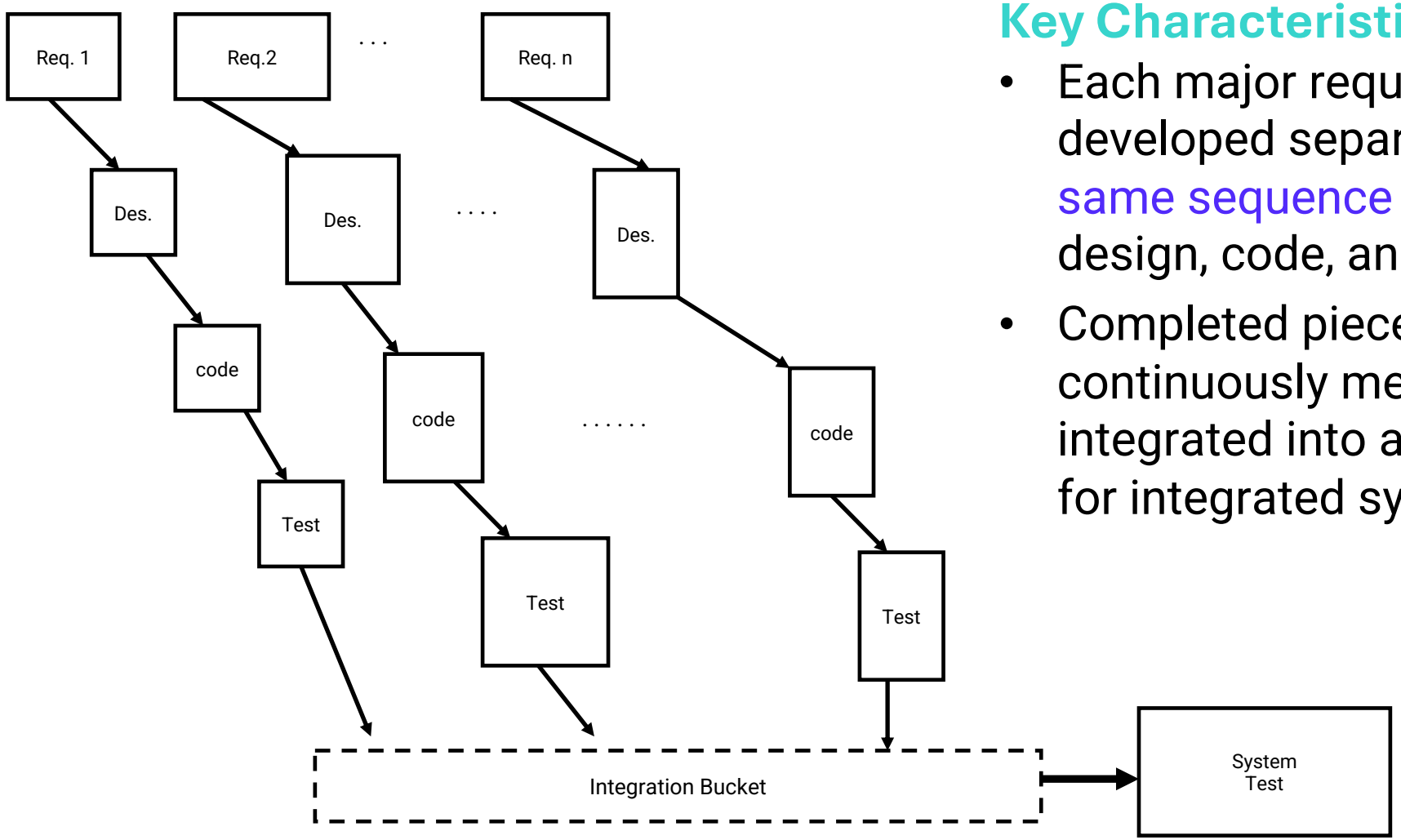
Pros

- Simple and easy to understand
- Well-suited for small projects with clear & fixed requirements
- Strong documentation supports long-term maintenance

Cons

- Lengthy development time
- Inflexible to changes once development begins
- Late discovery of issues (testing comes after coding)
- Poor fit for projects with evolving requirements or high uncertainty

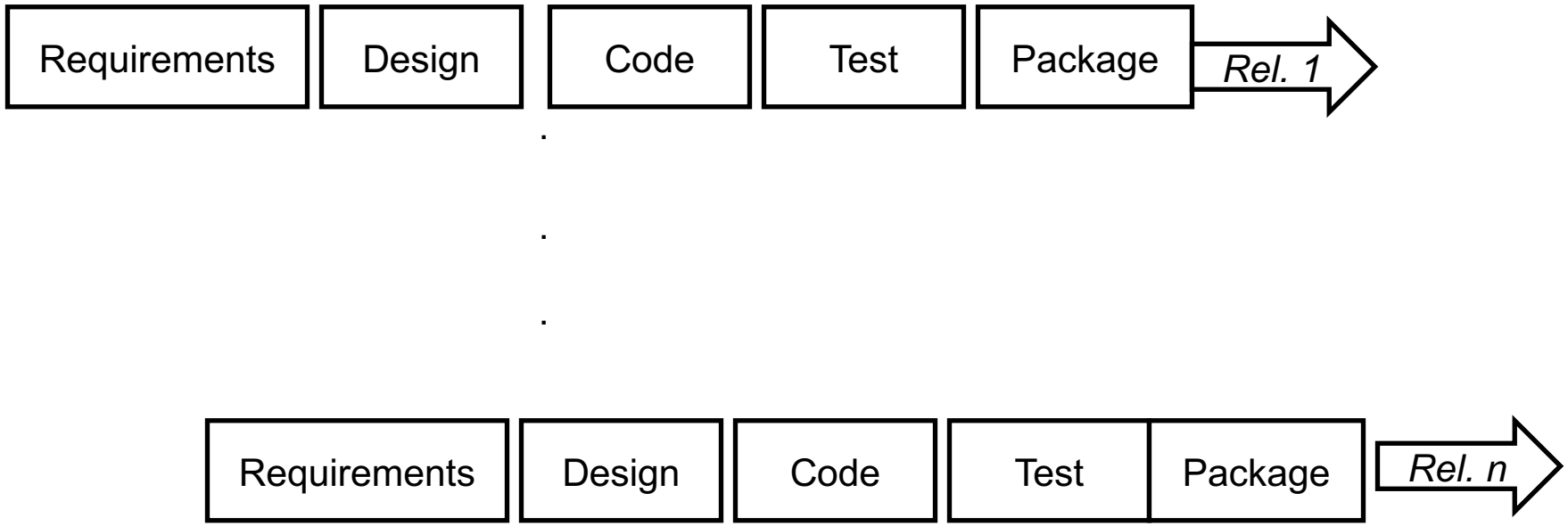
Incremental Model



Key Characteristics:

- Each major requirement/item is developed separately through the same sequence of requirement, design, code, and unit test
- Completed pieces are continuously merged and integrated into a common bucket for integrated system test

Incremental Model B - Multiple Release



Key Characteristic:

- Each small set of requirements is developed, packaged, and released in a multiple release fashion

Pros and Cons: Incremental Model

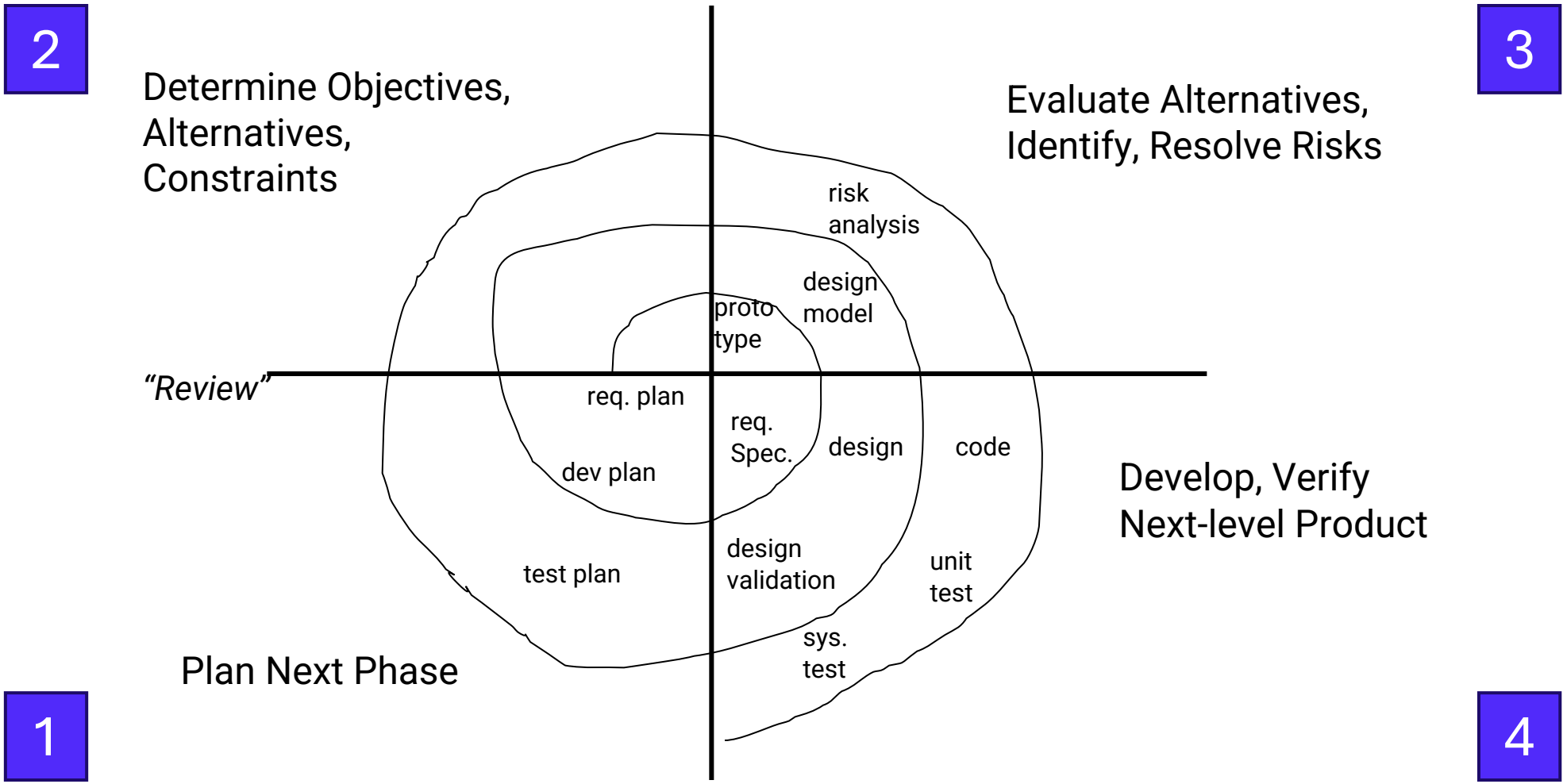
Pros

- Divide and conquer
- Suitable for a very large complex system

Cons

- Parallel execution
- May require more management efforts
- Integration challenges

Spiral Model



Pros and Cons: Spiral Model

Pros

- Require prototyping and modeling
- Iterative and evolutionary approach to all activities based on the number of risks involved
- Can revise things developed in an earlier increment

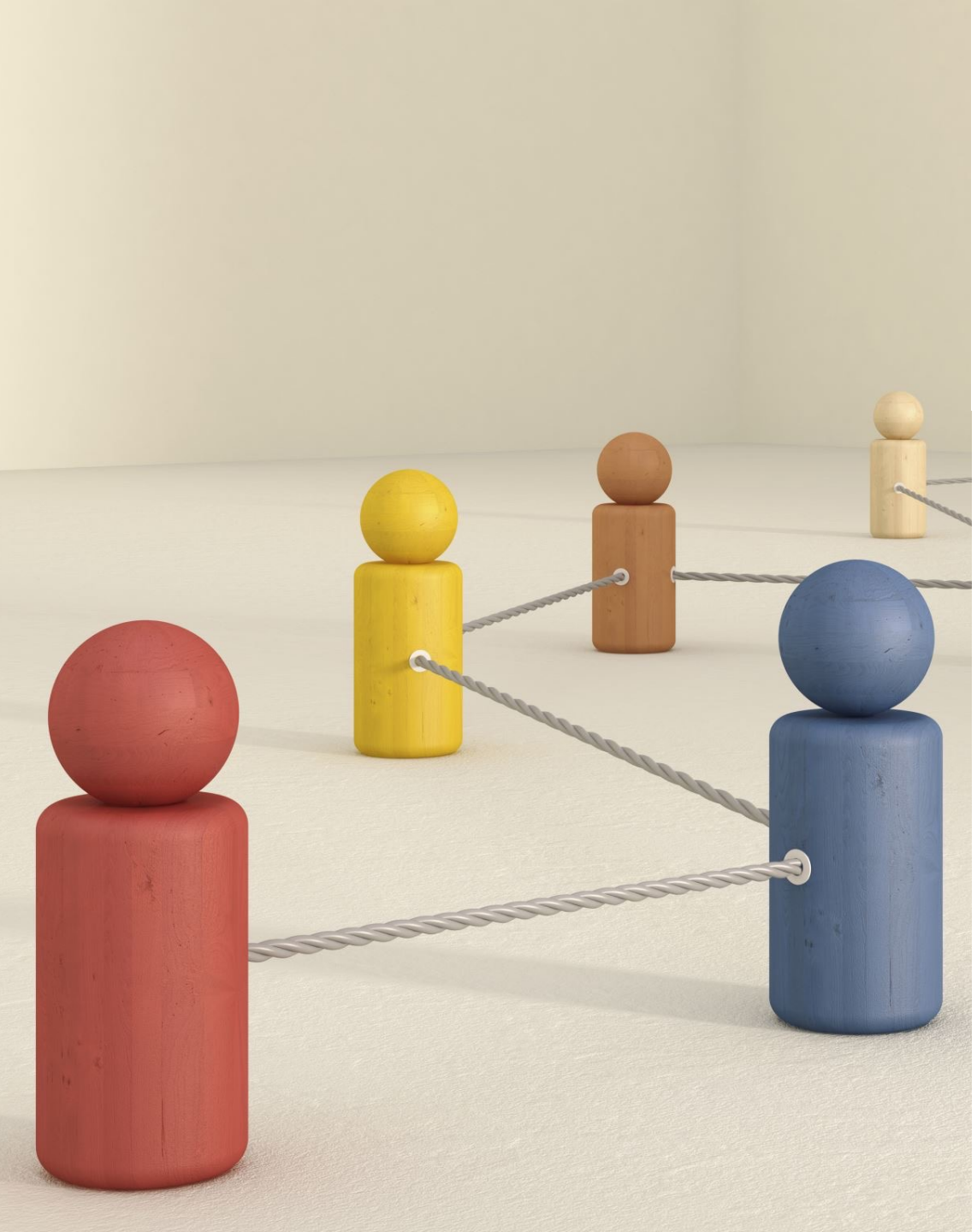
Cons

- Require expertise in risk assessment

Agile Model

- Agile Manifesto
  - *Individuals and interactions* over processes and tools
  - *Working software* over comprehensive documentation
  - *Customer collaboration* over contract negotiation
  - *Responding to change* over following a plan
- Examples:
  - Scrum
  - Extreme Programming (XP)
  - Crystal Clear/Orange
  - RUP!
  - Microsoft Solutions Framework

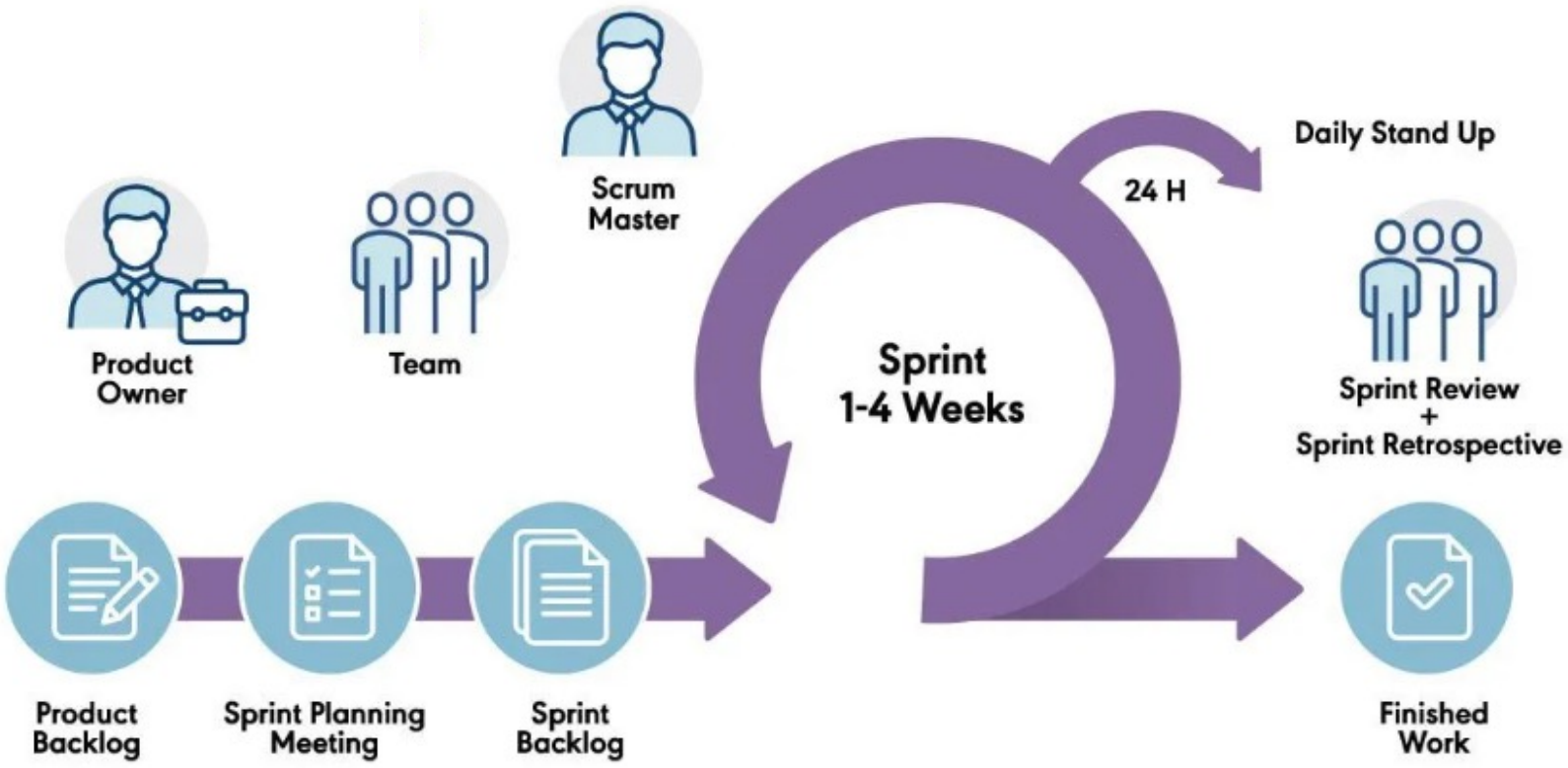
Reference: http://agilemanifesto.org



Agile Processes

- Family of software development methodologies
- Short releases and iterations
- Incremental design
- User involvement
- Minimal documentation
- Informal communications
- Change

Scrum



Key Characteristics:

- Iterative development
- Incremental delivery
- Customer collaboration
- Responding to change
- Cross-functional teams
- Minimal documentation

Reference: http://agilemanifesto.org; https://www.pm-partners.com.au/insights/the-agile-journey-a-scrum-overview/

Pros and Cons: Agile Model

Pros

- High flexibility and adaptability
- Faster delivery of functional software
- Greater customer satisfaction through regular feedback
- Early detection of problems

Cons

- Requires experienced and disciplined teams
- Less predictable costs and timelines
- Can be challenging for large, fixed-scope projects
- Documentation may be insufficient if not properly managed

Agile vs. Traditional

	Agile	Traditional / Heavy
Requirements	• Assumes change • Informal requirements • Constant user interaction	• Assumes no change • Complete, detailed, formal requirements document
Design	• Informal • Iterative	• Formal • Upfront
User involvement	• Crucial • Frequent	• Beginning (Requirements) • End (Acceptance testing)
Documentation	• Minimal, only as needed • Source code	• heavy, formal documents
Communication	• Informally • Throughout the project	• Documents • Formal memos and meetings
Complexity	• Low	• High
Overhead	• Low	• High

Case Study: Spotify's "Squad" Model

- Wanted to launch new features quickly in a competitive market
- Adopted an Agile approach with a twist:
  - **Squads:** small, autonomous, cross-functional teams owning a specific feature area
  - **Tribes**: groups of related squads for shared direction
  - Heavy emphasis on **continuous delivery**, A/B testing, and rapid feedback

Case Study: Spotify's "Squad" Model

- Why it worked?
  - **Fast iteration** - features went from concept to release in weeks
  - **Autonomy** - squads could adapt their work without waiting for top-down approval
  - **Data-driven decisions** - real-time user analytics informed priorities

Case Study: HealthCare.gov

- Wanted to launched **HealthCare.gov** as a central portal for Americans to sign up for health insurance
- Claimed to use an Agile approach:
  - Multiple contractors working on different modules
  - Frequent meetings and progress tracking

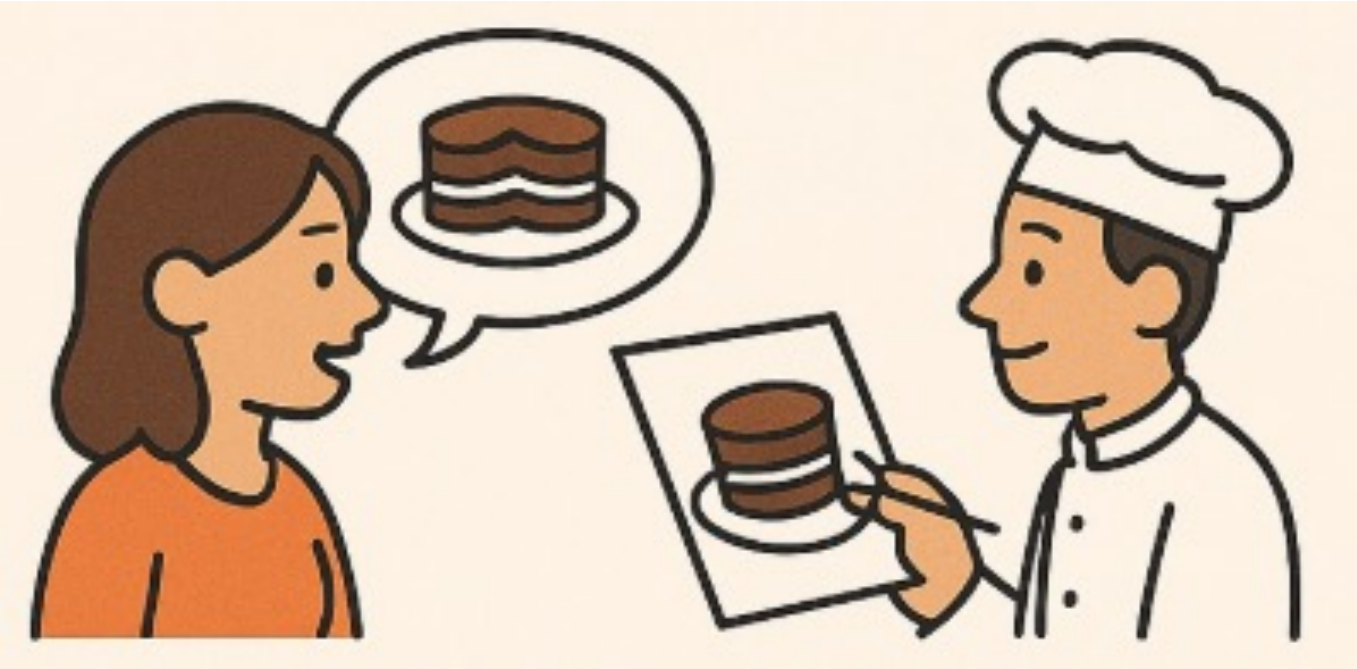
Case Study: HealthCare.gov

- Why it failed?
- **Fake Agile** - they kept rigid deadlines and massive scope like Waterfall, but without the coordination discipline
- **Poor integration** - components didn’t work together when assembled
- **Late testing** - system testing happened weeks before launch
- **No real user feedback cycles** - public couldn’t try it until the official launch day

Key comparison

Factor	Spotify success	HealthCare.gov failure
Iteration & Delivery		
User Feedback		
Team Autonomy		
Integration Testing		

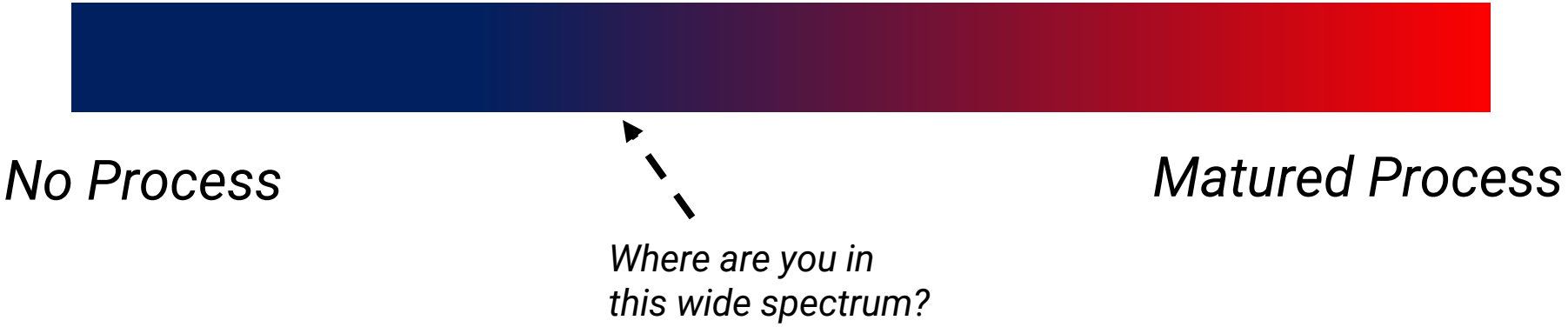
Exercise 1: Baking a Cake - Waterfall vs. Agile



Exercise 2: Making a Pizza - Waterfall vs. Agile

Process Assessment

- How mature is your software engineering organization and do you need to improve?
- ISO (ISO 9000 series) and SEI (at Carnegie Mellon) are two leading organizations that help in the **process assessment.**



Online resources

- Software development lifecycle
  - https://www.youtube.com/watch?v=Fi3_BjVzpqk&pp=ygUeU29mdHdhcmUgRGV2ZWxvcG1lbnQgTGlmZWN5Y2xl
  - https://www.youtube.com/watch?v=i-QyW8D3ei0
- Agile methodology
  - https://www.youtube.com/watch?v=8eVXTyIZ1Hs
- Scrum
  - https://www.youtube.com/watch?v=XU0llRltyFM